

Marc Zgheib

EDUCATION

MASTERS OF SCIENCE IN DESIGN ENGINEERING

Imperial College – London, United Kingdom

DYSON SCHOOL OF DESIGN ENGINEERING

September 2025 – September 2026

- Received the prestigious **fully funded CMA CGM scholarship**, given to only 6 students in 2025.

BACHELOR OF ENGINEERING IN MECHANICAL ENGINEERING

Lebanese American University (LAU) – Byblos, Lebanon

September 2021 – June 2025

- Granted the excellence **fully funded HES scholarship under the USAID Scholarship Program (USP VIII)** (2% acceptance rate).
- Cumulative GPA of 3.90/4.0.
- Consistently placed on the Dean's High Distinction List.
- Completed the University's Honors Program.
- Received the Young Researcher Award - 2025

EXPERIENCE

Internship at Unipak Tissue Mill – Member of INDEVCO Group

(Mechanical Engineering Trainee in Maintenance and Powerplant Departments)

June 2024 – August 2024

- Collaborated with experienced engineers and technicians in both departments to perform routine and preventive maintenance, troubleshoot and repair equipment.
- Participated in implementation of maintenance schedules, overhaul of engine components, and inspections of electrical systems, developing expertise in safety protocols, diagnostic tools, and industry standards.

Internship at Association of Energy Engineers (AEE), Lebanon

July 2022 – August 2022

- Worked on-site with 5 engineering students on power generation plans for a project in Dammour, created simulations, and worked on all the calculations needed for the turbines.
- Attended a rich selection of masterclasses delivered by experts on topics related to energy, sustainability, and personal development.

CERTIFICATES & EXTRACURRICULARS

- Received the **SolidWorks accreditation** of the Associate Mechanical Design (October 2022).
- Received the **Young Researcher Award – 2025** for my publication in the Energy Conversion and Management journal, titled "[Machine Learning Applications in Optimizing the Performance of Darrieus Wind Turbines.](#)"
- Top 10 local finalist** in the L'Oréal Brandstorm (March 2024).
- More than **450 total hours** of volunteering throughout university enrollment (essential for HES/USAID scholarship).
- Wrote articles about tech & innovation for the HELM monthly magazine (2021-present).
- Vice President** of the Astronomy Club at LAU (2021-2022).
- Member of the Robotics and ASME clubs at LAU (2021-present).
- Undergraduate Research** Projects in thermodynamics and energy fields with Dr. Amne el Cheikh.
- Completed the online course from the **University of Geneva**: "Particle Physics: an introduction" (September 2023).
- Completed the online course from the **Georgia Institute of Technology**: "Machine Design" (January 2024).
- Facilitated** "Power of space" workshop for LEE program (2022).
- Selected as **HES mentor** for 2 consecutive years (2022-present).
- Usher for "Engineering Summer Camp" at LAU (June-July 2023).
- Model United Nations (MUN) at LAU (2015).
- Classical and Acoustic Guitar player.
- Competitive League Padel Player (Imperial College).
- Passionate about hiking, backpacking, camping, and the outdoors.

CONTACT

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in <http://bit.ly/4cSKTl0>

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📍 London, UK

📍 Kesrwan, Lebanon

SKILLS

Language skills

- English (Native Lvl.)
- French (Native Lvl.)
- Arabic (Native Lvl.)

Technical skills

- SolidWorks
- MATLAB
- LabVIEW
- Java
- AutoCAD
- KiCAD
- Keyshot
- Max/MSP
- Python
- Arduino (C++)
- Blender
- ANSYS
- EES
- Homer
- Primavera
- Carrier HAP

Soft skills

- Problem-Solving
- Leadership
- Critical Thinking
- Communications
- Creativity
- Team Collaboration
- Project Management
- Adaptability

DESIGN PROJECTS

E-PORTFOLIO link:

<https://marczgheib1.github.io/E-Portfolio/>

- Dual axis, IoT connected, desktop laser turret with auto face tracking and emotion detection models + SQL and data analytics user interface
- Real-time kaleidoscope video-to-music translation device with custom computer vision pipeline, Raspberry Pi/Arduino hardware stack, and real-time Max/MSP audio synthesis.
- Wearable wrist heat therapy device for arthritis relief in elderly users, featuring flexible PCB heating, Arduino Nano BLE control, haptic feedback, and a simple LED interface integrated into a neoprene strap.
- Handheld, gamified, grip/wrist joint training device for the elderly and people with dexterity problems.
- Wearable gait retraining system using pressure-sensing insoles
- 3-wheeled novel e-scooter, a multi-terrain first response micro-mobility solution
- Motorized sun-tracking patio umbrella (2.5m steel structure)
- Bluetooth rover with obstacle detection